A MORPHISM WHICH IS FLAT AWAY FROM A RELATIVE EFFECTIVE CARTIER DIVISOR IS FLAT EVERYWHERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-michael-artin is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 4141-4146 and proof span 4148-4168 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. Reread under the 2026-10-14 prefilter rule: this candidate is 817 bytes over 21 proof lines and the prefilter matched the string 'Some details omitted.' in the last line of the proof. On rereading, the proof gives the algebra reduction, two explicit short exact sequences and the induction that makes the quotients flat, and the omitted details are the routine bookkeeping of the translation, so the candidate was accepted rather than skipped; the omission is recorded here and is not counted as a gap in the statement.

Editorial note. Tag self-check per LANE_BRIEF 10.5 (5): the label lemma-michael-artin resolves in the project's own tag list to canonical tag 062W (line 7810 of the index, full label divisors-lemma-michael-artin). Since reference/public-items.jsonl holds no row whose primary source_file is divisors.tex, the same-chapter public tag id set is empty and the intersection with the resolved tag id is empty, so the candidate passes.

Editorial note. The gate was run as: python operations/agent-authoring/tools/precard_gate.py –source divisors.tex –label lemma-michael-artin –start 4141 –end 4146 –proof-start 4148 –proof-end 4168 –local-id 34089, giving RESULT=OK with the label found inside the statement span, disjoint statement and proof spans, exactly one proof environment between them, distinct statement and proof text, no public interval row for this chapter and occupancy FREE for divisors.tex 4141 4168. Spans were recomputed live by locating the label in the pinned chapter file, and the located text matched the pool byte counts (statement 192 characters, proof 674 characters after whitespace removal).

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The algebra lemma that a module in a short exact sequence with the two outer terms flat is flat, cited from the algebra chapter. Cited at proof lines 4157-4159. Applied to the displayed sequence with the localization of B in the middle and the colimit of the quotients on the right, it gives flatness of B from flatness of the two outer terms.
- The algebra lemma that a filtered colimit of flat modules is flat, cited from the algebra chapter. Cited at proof lines 4159-4161. Shows that the third term of the displayed sequence is flat over A once each quotient B/h^nB is flat.
- The definition of a relative effective Cartier divisor: locally it is cut out by a nonzerodivisor on the source, cited from the same chapter. Statement lines 4143-4145 and proof lines 4148-4152. Supplies the element h which is a nonzerodivisor in the algebra translation, which is what makes the displayed sequence exact.

Source reference. Original lemma 05UT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7527–7527. Same-chapter reference used in the retained author text at source line(s) 4161, 7841. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 00HM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 648–648. Same-chapter reference used in the retained author text at source line(s) 4013, 4037, 4159. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $D \subset X$ be a relative effective Cartier divisor on X/S. If f is flat at all points of $X \setminus D$, then f is flat.*

Proof. This translates into the following algebra fact: Let $A \to B$ be a ring map and $h \in B$. Assume h is a nonzerodivisor, that B/hB is flat over A, and that the localization B_h is flat over A. Then B is flat over A. The reason is that we have a short exact sequence

$$0 \to B \to B_h \to \operatorname{colim}_n (1/h^n)B/B \to 0$$

and that the second and third terms are flat over A, which implies that B is flat over A (see Algebra, Lemma 00HM). Note that a filtered colimit of flat modules is flat (see Algebra, Lemma 05UT) and that by induction on n each $(1/h^n)B/B \cong B/h^nB$ is flat over A since it fits into the short exact sequence

$$0 \to B/h^{n-1}B \xrightarrow{h} B/h^nB \to B/hB \to 0$$

Some details omitted. □